

RS004 Week 9 — English Worksheet

Topic: Level Design (Data, Respawn, Finish) · **Course:** Platformer Game · **Time:** about 45 minutes

This week you separate your **level data** from your game code. A `build_level()` function turns a list of numbers into platforms, the player **respawns** at the start if it falls off, and a **finish** marker says "you cleared it!".

Keep these words handy: **level data**, `build_level()`, **respawn / start position**, **finish**, **cleared**.

1 · Predict 🧠

Read each snippet. Before you run it, write what you think will happen.

```
LEVEL_1 = [
    (0, 550, 800, 50),
    (150, 450, 120, 20),
]
def build_level(level_data):
    return [pygame.Rect(*p) for p in level_data]
```

What does `build_level(LEVEL_1)` give back? What does `*p` do?


```
if player.y > HEIGHT:
    player.x = start_x
    player.y = start_y
    velocity_y = 0
```

The player falls below the bottom of the screen. What happens to it?


```
if player.colliderect(finish):  
    cleared = True
```

Once **cleared** is **True**, what should change about the game?

2 · Why Separate the Data? 🤔

We keep the level as a list of numbers (`LEVEL_1`) instead of writing each `pygame.Rect(...)` by hand in the loop.

Give one reason this makes it easier to build *many* levels later.

3 · Spot the Bug 🐛

Each snippet is broken. Rewrite it so it works, then explain why the original was wrong.

Bug A — `build_level` should return real platforms, but the game crashes saying it got tuples, not Rects.

```
def build_level(level_data):  
    return level_data
```

Hint: each tuple of numbers must be turned into a `pygame.Rect`.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — A player who falls off should respawn at the start. Right now it keeps falling forever and never comes back.

```
if player.y > HEIGHT:  
    velocity_y = 0
```

Hint: resetting velocity isn't enough — move the player back to its start spot.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — Touching the finish should clear the level once. Right now the "Level Clear!" text never shows.

```
finish = pygame.Rect(710, 180, 50, 40)
if player.colliderect(player):
    cleared = True
```

Hint: the player is being compared with *itself*, not the finish.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Shape a Level

Start from your working level loader.

1. Move one platform 60 pixels higher. Can the player still reach it with a jump? Note the result.

2. Move the finish marker somewhere harder to reach. Where did you put it?

3. **Stretch:** show a "Level Clear!" message on screen when `cleared` is true. What font size did you use?

5 · Build & Show 📷

Design **your own** level data. It should have a clear start, a path of platforms, and a finish marker. Make sure the whole level is reachable.

When it works, send a **photo or video** of a full run from start to finish, then explain what you did. Use these starters — write 4 to 6 sentences.

First, I kept my level as data so that ...

`build_level()` turns the numbers into ...

When the player falls off, it respawns by ...

My finish marker is at ... and the player reaches it by ...

One tricky moment was ...

If I had more time, I would ...

6 · Record Your Walkthrough

Take a video of a full clear of your level. Teach it to someone new. Try to use these words: **level data**, **build_level**, **respawn**, **finish**, **cleared**.

1. Show your level and point out the start and finish.
2. Fall off on purpose and show the respawn.
3. Read the respawn lines out loud and explain them.
4. Reach the finish and show the clear.

Write what you will say in your video. Plan it here first.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.